# Automated Wheatgrass Production System – Phase II (Digital Twin) Hardware Installation Manual

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# Table of contents

**1. Document Purpose and Scope** **4**
**2. Installation Overview** **5**
**3. Required Hardware, Tools, and Materials** **6**
**4. Recommended Placement and Mounting Guidance** **8**
**5. Detailed Installation Procedures** **9**
5.1 Power Down and Site Preparation 9
5.2 Mount the Raspberry Pi 9
5.3 Mount the ESP32 enclosure or board 9
5.4 Mount the BME280 sensor 9
5.5 Mount the TSL2591 light sensor 9
5.6 Install the water-flow sensor, if used 9
5.7 Connect wiring and power 10
5.8 Restore power and confirm boot 10
**6. Basic Troubleshooting During Installation** **11**
**7. Safety and Handling Notes** **12**

# 1. Document Purpose and Scope

This installation manual describes how to install the deployed monitoring hardware used in the Automated Wheatgrass Production System - Phase II. It focuses on mounting, placement, wiring, initial startup, and post-installation verification for the monitoring subsystem. The manual does not cover mechanical redesign of the conveyor system or advanced stretch-goal features that were outside the Phase II core scope.

# 2. Installation Overview

The installation brings together environmental sensors, a local ESP32 sensing node, and a Raspberry Pi backend so that environmental data can be collected, transmitted, and viewed through the monitoring interface. Correct sensor placement and protected electronics mounting are necessary to obtain valid readings and reliable operation.

| **Layer** | **Main Components** | **Role in Installation** | **Installation Relevance** |
|---|---|---|---|
| Sensing | BME280, TSL2591, YF-S201 | Mounted near the growth and irrigation areas to capture representative measurements | Requires suitable placement, moisture protection, and secure cable routing |
| Control | ESP32 development board | Reads sensors and transmits data to the backend over Wi-Fi | Must be mounted in a dry location with stable 5 V power and low connector strain |
| Network | Portable Wi-Fi / local LAN path | Provides communication between the ESP32 and Raspberry Pi | Requires 2.4 GHz connectivity and adequate signal before final placement |
| Backend | Raspberry Pi 4, Home Assistant, storage media | Receives, stores, and visualizes data after installation | Must remain accessible for power, rebooting, SD card service, and future maintenance |

*Table 1. Installation layers and functions.*

# 3. Required Hardware, Tools, and Materials

| Item | Purpose | Installation Notes |
|---|---|---|
| ESP32 development board | Reads sensors and transmits data to backend | Mount in a dry location; use stable 5 V power; requires 2.4 GHz Wi-Fi. |
| BME280 sensor | Measures temperature and humidity | Place near the growth area where airflow is representative, but avoid direct water droplets. |
| TSL2591 light sensor | Measures ambient light level | Mount where it sees representative grow light; avoid physical shading. |
| YF-S201 flow sensor (if installed) | Monitors water flow | Install inline with the water path and verify direction/orientation. |
| Raspberry Pi 4 | Runs Home Assistant / backend services | Install near stable power and network source; keep accessible for maintenance. |
| Portable Wi-Fi / local network | Connects ESP32 to Raspberry Pi | Use 2.4 GHz Wi-Fi and verify signal before finalizing sensor placement. |
| Basic installation tools | Supports mounting and cable management | Small screwdriver set, cable ties or adhesive clips, double-sided tape, screws, or brackets. |

| | | |
|---|---|---|
| Support items | Simplifies final setup and checks | 5 V power adapters, USB cables, label tape or marker, and a laptop for setup and verification. |

*Table 2. Hardware, tools, and materials required for installation.*

Note: Install the electronics indoors and away from direct water spray. Place the ESP32 and power connections in a protected enclosure or mounting location above the irrigation zone.

# 4. Recommended Placement and Mounting Guidance

- Mount the BME280 near the wheatgrass growth area so it measures representative chamber temperature and humidity.
- Do not place the BME280 where it is directly sprayed by irrigation mist or pressed against a heat source.
- Mount the TSL2591 so it receives representative light from the grow lighting, without being blocked by frames, trays, or cable bundles.
- Place the ESP32 in a dry, protected area above the irrigation zone to reduce moisture exposure.
- Route cables along the frame of the machine and secure them so they do not interfere with trays, conveyors, pumps, or lighting.
- If the YF-S201 flow sensor is used, install it inline with the water path and ensure the tubing is firmly seated before pressurizing the system.

# 5. Detailed Installation Procedures

## 5.1 Power Down and Site Preparation

- Turn off or unplug the Raspberry Pi, ESP32 power adapter, and any nearby hardware that may interfere with the installation.
- Confirm the work area is dry before handling electronics or routing cables.

## 5.2 Mount the Raspberry Pi

- Place the Raspberry Pi in a stable indoor location near power and network access.
- Keep the Raspberry Pi accessible for SD card service, cable inspection, and rebooting.
- Connect the Raspberry Pi power supply, but leave it powered off until the sensing node is installed.

## 5.3 Mount the ESP32 enclosure or board

- Install the ESP32 in a dry location above the irrigation zone or inside a protective enclosure.
- Leave enough slack in the sensor and power wiring to avoid connector strain.
- Avoid locations with condensation, standing water, or direct contact with wet surfaces.

## 5.4 Mount the BME280 sensor

- Secure the BME280 near the growth area where the reading reflects the actual chamber microclimate.
- Keep the sensor away from direct misting, puddles, or surfaces that trap heat.
- Ensure the cable is secured and does not hang into moving or wet areas.

## 5.5 Mount the TSL2591 light sensor

- Place the light sensor where it can see representative illumination from the wheatgrass light source.
- Do not mount it under structural shadows or behind opaque covers.
- Secure the cable so the sensor orientation remains stable after installation.

## 5.6 Install the water-flow sensor, if used

- Insert the YF-S201 inline with the water path according to the selected plumbing arrangement.
- Tighten fittings and support the tubing so the sensor body is not carrying hose tension.
- Check the flow direction marking before resuming water flow.

## 5.7 Connect wiring and power

- Connect the sensors to the ESP32 according to the project wiring configuration.
- Use stable 5 V power for the ESP32 and Raspberry Pi.
- Bundle and secure all cables to prevent motion, snagging, or drip contact.

## 5.8 Restore power and confirm boot

- Power on the Raspberry Pi and allow the backend services to boot.
- Power on the ESP32 and confirm it starts normally.
- Verify that the ESP32 joins the configured 2.4 GHz Wi-Fi network.

# 6. Basic Troubleshooting During Installation

| **Issue** | **Likely Cause** | **Immediate Action** |
|---|---|---|
| No sensor readings | Loose wiring, incorrect I2C connection, or sensor not powered | Recheck connectors, power, and the configured sensor wiring. |
| ESP32 offline | Wrong Wi-Fi credentials, weak signal, or insufficient power | Verify 2.4 GHz network, move device if needed, and confirm stable 5 V supply. |
| Humidity or temperature looks wrong | BME280 placed too close to water, heat, or dead-air zone | Relocate sensor to a more representative chamber location. |
| Light value too low or unstable | TSL2591 shaded or poorly oriented | Reposition the sensor so it sees the grow light directly. |
| Flow sensor not responding | Incorrect orientation, poor wiring, or no real flow | Check inline direction, connector seating, and tubing path. |

*Table 3. Common installation issues and immediate actions.*

# 7. Safety and Handling Notes

- Do not work on the electronics with wet hands or while the installation area is actively being sprayed.
- Keep all low-voltage electronics away from direct water contact.
- Do not route cables where they can be pinched by trays, doors, or moving assemblies.
- Do not block ventilation openings of the Raspberry Pi or enclosures.
- Label cables after installation to simplify future maintenance.